MATH MA’AM ACADEMY

Where Every Subject Clicks.

STD 5 MATHEMATICS – CHAPTER 5: ANGLES

Learning objectives: Point, line and line segment • Angles • Types of angles

5.1 POINT, LINE AND LINE SEGMENT

Point: An exact position or location. It is represented by a dot and named using a capital letter, such as point P.

Line: A straight path that extends endlessly in both directions. It has no endpoints.

A line extends endlessly in both directions

Line segment: A part of a line with two endpoints and a definite length. It can be measured.

A line segment has two endpoints

Ray: A part of a line that starts at one fixed endpoint and extends endlessly in one direction.

A ray has one endpoint and extends in one direction

Types of lines

• **Intersecting lines:** Two lines that cross and meet at a common point, called the point of intersection.

• **Parallel lines:** Two lines in the same plane that never meet, even when extended in either direction.

Remember: A line has no endpoints; a ray has one endpoint; a line segment has two endpoints.

5.2 ANGLES

An **angle** is formed when two rays or line segments meet at a common endpoint. It represents the amount of turn between the two arms.

Parts of an angle

• **Vertex:** The common endpoint where the two arms meet.

• **Arms:** The two rays that form the angle.

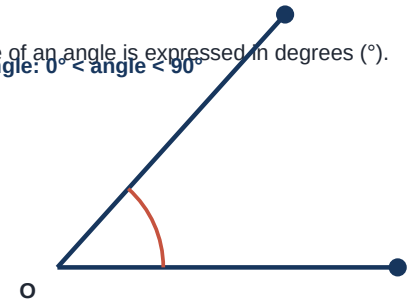
• **Angle symbol:** The symbol ∠ is used to represent an angle.

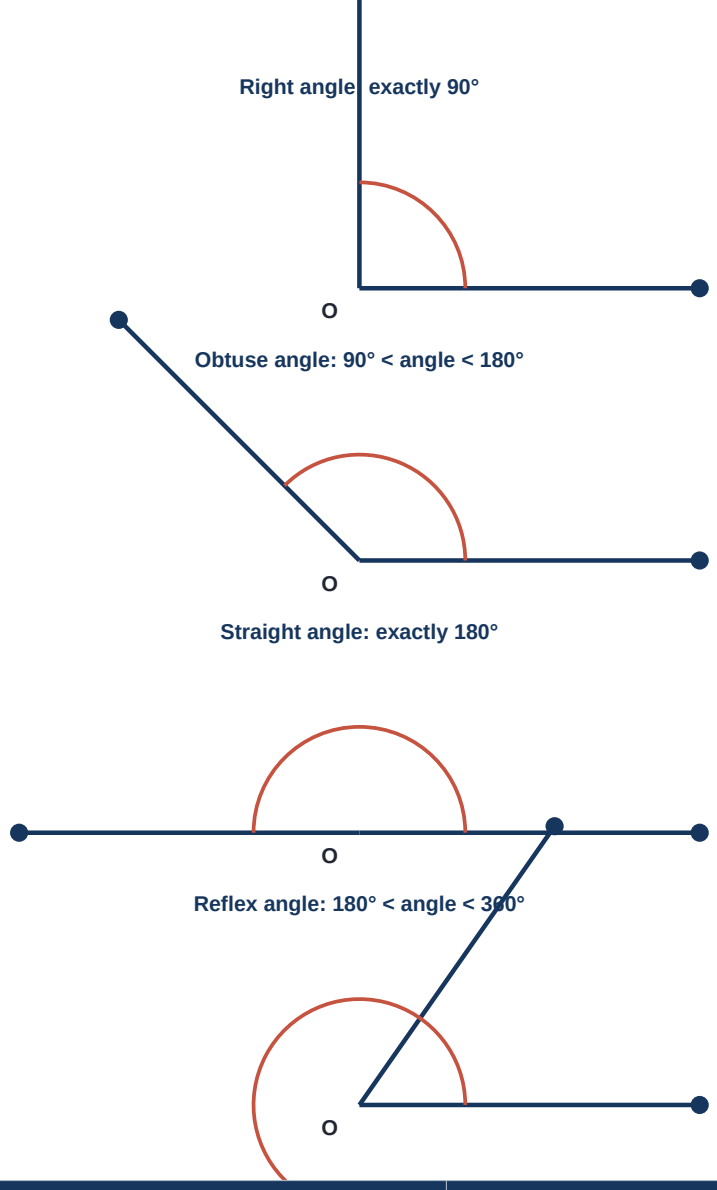
How to name an angle: In ∠ABC, the middle letter B is the vertex. The arms are BA and BC.

5.3 TYPES OF ANGLES

Angles are classified according to their measure. The measure of an angle is expressed in degrees (°).

Acute angle: 0° < angle < 90°





Type	Measure	Identification
Acute	More than 0° but less than 90°	Smaller than a right angle
Right	Exactly 90°	A square corner
Obtuse	More than 90° but less than 180°	Between a right and straight angle
Straight	Exactly 180°	Forms a straight line
Reflex	More than 180° but less than 360°	The larger turn around the outside

Important: 90° is a right angle, not acute or obtuse. 180° is a straight angle. 360° is one complete turn.

ANGLES AS TURNS

Turn	Angle measure	Example
Quarter turn	90°	One right-angle turn
Half turn	180°	Facing the opposite direction
Three-quarter turn	270°	Three right-angle turns
Complete turn	360°	Returning to the starting direction

- One complete turn = 4 right angles = 360°.
- One half turn = 2 right angles = 180°.
- One quarter turn = 1 right angle = 90°.

READING ANGLES FORMED BY CLOCK HANDS

The hands of a clock form angles at the centre. Unless the question asks for a reflex angle, identify the smaller angle between the hands.

Clock time	Smaller angle	Type
3:00	90°	Right angle
6:00	180°	Straight angle
2:00	60°	Acute angle

Clock time	Smaller angle	Type
4:00	120°	Obtuse angle
9:00	90°	Right angle

Clock tip: At an exact hour, the minute hand points to 12. Each number interval on the clock represents 30°.

ANGLES IN GEOMETRICAL FIGURES

• A triangle has 3 vertices and 3 interior angles.

• A quadrilateral has 4 vertices and 4 interior angles.

• A pentagon has 5 vertices and 5 interior angles.

• A hexagon has 6 vertices and 6 interior angles.

• In a simple closed polygon, each corner is a vertex and has one interior angle.

To count angles in an irregular or concave outline, trace its boundary carefully and count each corner once. Do not count points that lie along a straight side as additional corners.

OLYMPIAD QUICK REVISION

Question clue	Correct concept
A dot showing a position	Point
Extends endlessly both ways	Line
Two endpoints	Line segment
One endpoint and one direction	Ray
Lines that cross	Intersecting lines
Lines that never meet	Parallel lines
Common endpoint of two arms	Vertex
Less than 90°	Acute angle
Exactly 90°	Right angle
Between 90° and 180°	Obtuse angle
Exactly 180°	Straight angle
Between 180° and 360°	Reflex angle

Common mistakes to avoid:
• Do not confuse a line with a line segment.
• The middle letter in a three-letter angle name gives the vertex.
• An obtuse angle is smaller than 180°; a reflex angle is larger than 180°.
• A right angle is exactly 90°.

IMO practice focus: Identify lines and rays from diagrams, locate vertices and arms, classify angles by size, recognise angles in clock hands, count corners in irregular shapes, and compare angles with 90°, 180° and 360°.

ALL THE BEST